

Did the COVID-19 Housing Boom Restore Great Recession Losses?

Heterogeneous Effects Across the Income Distribution and Local Housing Markets

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Abstract

Using IPUMS USA American Community Survey microdata for owner-occupied households, I conducted two complementary analyses. At the state level, I compared housing value changes across low-income (bottom 20%), middle-income (20–80%), and high-income (top 20%) homeowners. While nearly all state-income cells showed nominal recovery by 2021, state-level regressions provided overall weak evidence of a systematic “payback”; estimated slopes were small in magnitude except for middle-income group, with very low explanatory power and were at best marginally significant, indicating that states with larger Great Recession losses did not consistently experience larger COVID-era gains. At the PUMA level, I identified the local markets most “decimated” by the Great Recession and found that 78% of these areas had recovered to their 2007 home values by 2021, compared with 90% of other PUMAs, leaving a persistent 12 percentage-point recovery gap. Taken together, the results suggested that although most housing markets eventually surpassed pre-crisis levels in nominal terms, the COVID boom did not equitably and reliably target the places or income groups that were hit hardest, implying that sequential asset-price booms is an unreliable mechanism for repairing the geographic and distributional damage of housing crises.

1. Introduction

The Great Recession of 2007–2009 was, at its core, a housing crisis. The collapse of house prices destroyed trillions of dollars in household wealth, triggering a cascade of reduced consumption, rising foreclosures, and severe employment losses. More than a decade later, the COVID-19 pandemic produced a significant

reversal in housing markets. Historically low interest rates, a surge in remote work, and constrained supply drove home prices to record highs in many parts of the United States between 2019 and 2021.

This paper asks a simple but important question: Did the COVID-era housing boom restore the housing value losses from the Great Recession, and did this recovery differ across the income distribution? The question matters for two very important reasons. Firstly, housing is the dominant asset on the balance sheet of most American households, particularly those in the middle of the wealth distribution. If housing values took more than a decade to recover, the implied wealth losses were persistent, with consequences for consumption, retirement planning and generational wealth transfers. Secondly, if the recovery was uneven across income groups, then two successive housing shocks; both negative and positive; may have widened wealth inequality rather than restoring it.

The foundation for this project draws on two significantly important papers we studied in the course. Professors Mian and Sufi's research on household balance sheets and the Great Recession demonstrates that areas experiencing larger declines in housing net worth suffered disproportionately larger drops in spending and employment, with effects concentrated among highly leveraged homeowners. Their work establishes that the geographic distribution of housing shocks was a first-order determinant of the severity of the great recession. Professor Campbell's (2006) work on household finance provides the complementary insight that housing constitutes the largest component of wealth for middle-class households, who hold relatively undiversified portfolios compared to the wealthy. Together, these perspectives imply that housing price shocks have highly unequal effects across the income distribution and that subsequent price recoveries may also be distributed unevenly.

My extension connects these two lines of research by linking the Great Recession crisis to the COVID-era boom. Rather than examining how housing shocks affect consumption or employment, I focused on whether the housing values themselves recovered and whether this recovery was shared across income groups. The analysis proceeds in two parts. In the first part, at the state level, I classified homeowners into bottom 20%, middle 60%, and top 20% income groups and tested whether COVID-era housing gains were systematically larger in states that experienced worse Great Recession losses, separately by income group. In the second part, at the PUMA level (Public Use Microdata Area, approximately 100,000 people), I identified the local markets that were most severely damaged during the Great Recession and determined what fraction had fully recovered by 2021.

2. Related Literature

This project is situated at the intersection of two foundational literatures in household and macroeconomic finance.

The first concerns the real economic consequences of housing wealth shocks. In "What Explains the 2007–2009 Drop in Employment?" (Econometrica, 2014), Professors Atif Mian and Amir Sufi demonstrate that the collapse in housing net worth, defined as the change in house prices scaled by pre-crisis housing wealth relative to total household net worth, was a primary driver of the sharp decline in U.S. employment during the Great Recession. Using cross sectional variation across 944 U.S. counties, they show that areas with larger housing net worth declines experienced significantly greater job losses in non-tradable sectors such as retail and restaurants, consistent with a household balance sheet channel in which falling home values suppress local consumer demand. Crucially, they find that this relationship is not explained by industry specific supply side shocks, exposure to the construction sector, policy induced business uncertainty, or contemporaneous credit supply tightening. They further document that there is no significant offsetting expansion of tradable-sector employment in the hardest-hit counties, little evidence of downward nominal wage adjustment, and negligible net out-migration from severely affected areas, which suggests that the local employment consequences of the housing net worth shock were both large and persistent.

The second literature concerns the distribution of wealth and the composition of household portfolios. In his American Finance Association Presidential Address, "Household Finance" (The Journal of Finance, 2006), Professor John Y. Campbell provides a systematic framework for understanding how households invest versus how normative theory prescribes they should. He documents that housing is the dominant asset class for households in the middle of the wealth distribution, while wealthier households hold more diversified portfolios with larger shares of financial assets. This compositional difference means that a housing price shock has a disproportionate impact on middle-class and lower-income homeowners relative to the wealthy. Combined with evidence from the Survey of Consumer Finances and related studies showing that wealth growth has diverged across the income distribution since the Great Recession, this work suggests that a housing boom may not equalize the distributional damage of a prior bust, even if average prices recover.

My contribution is to empirically test whether the COVID housing boom provided a systematic “payback” of Great Recession losses and whether this payback differed by income group. By combining PUMA-level geographic analysis with state-level distributional analysis using a single consistent data source (ACS microdata via IPUMS), the paper directly links the two episodes in a way that draws on both the Mian-Sufi geographic shock framework and the Campbell portfolio composition perspective.

3. Data and Sample Construction

Data source: I used public-use microdata from IPUMS USA, drawing on the American Community Survey (ACS) 1-year samples for 2007, 2009, 2019, and 2021. The 2007 and 2009 samples indicate the Great Recession period, while 2019 and 2021 indicate the COVID-era housing boom.

Sample restrictions: The sample is restricted to owner-occupied housing units (OWNERSHP = 1) in normal households (GQ codes 1 and 2, capturing both the 1970 and 1990 household definitions and excluding group quarters). I dropped observations with missing or not-in-universe values for household income (HHINCOME = 9,999,999) and housing value (VALUEH codes 0, 9,999,998, and 9,999,999). After these restrictions, the working sample contained 8,955,228 household-year observations across 51 states (including the District of Columbia) and 1,185 unique PUMA codes.

Key variables: The primary outcome variable is VALUEH, the self-reported value of the housing unit. The income variable is HHINCOME, total household income. Geographic identifiers include STATEFIP (state FIPS code) and PUMA (Public Use Microdata Area). All weighted computations use HHWT, the household weight provided by IPUMS. Since PUMA codes are state-dependent, I constructed a unique PUMA identifier by concatenating STATEFIP and PUMA, yielding 3,344 unique state-PUMA combinations and 1,076 with complete data across all four years.

Income group classification: For the state-level analysis, I computed weighted 20th and 80th percentile thresholds of HHINCOME within each state-year cell and classified households into three groups: low income (bottom 20%), middle income (20–80%), and high income (top 20%). Using state-specific percentiles ensures that income group assignment reflects local economic conditions as a household earning \$80,000 may be classified as high-income in Mississippi but middle-income in California.

Winsorization: To limit the influence of extreme housing values (very high-value properties or data anomalies), I winsorized VALUEH at the 1st and 99th percentiles within each geographic-unit-by-year-by-income-group cell before computing weighted means. This preserves the full sample size while reducing the impact of outliers on group averages.

4. Empirical Methodology

The empirical analysis is organized in two complementary parts, each using a different geographic unit and addressing a distinct aspect of the research question.

4.1 Part 1: State-Level Income Group Analysis

For each income group, at the state level, I computed weighted mean housing values for each state in 2007, 2009, 2019, and 2021 and calculated percentage changes during the Great Recession period (2007–2009) and the COVID boom period (2019–2021). The core regression specification, estimated separately for each income group g , is:

$$\% \Delta \text{Value}_{2007-2009, s} = \alpha + \beta \times \% \Delta \text{Value}_{2019-2021, s} + \varepsilon_s$$

where s indexes states. A negative and statistically significant coefficient β would indicate that states with worse Great Recession losses tended to have experienced larger COVID-era housing gains.

4.2 Part 2: PUMA-Level Recovery of Decimated Markets

For each PUMA with complete data across all four years, I computed the weighted mean housing value in each year and calculated the percentage changes for the mentioned two periods. I then define “decimated” PUMAs as those in the bottom quartile of the 2007–2009 percentage change distribution, that is, the 25% of local markets that experienced the largest housing value declines during the Great Recession. The cutoff for decimation is a percentage decline of 9.67% or worse. A PUMA is classified as “recovered” if its 2021 weighted mean housing value equals or exceeds its 2007 level. I then compare recovery rates between decimated and non-decimated PUMAs and visualize the joint distribution of Great Recession and COVID-era changes using scatter plots with separate regression lines.

5. Results

5.1 State-Level: Recovery Across Income Groups

For each income group, Table 1 summarizes the state-level recovery statistics by income group. All 51 state-income cells for high-income homeowners and all 51 for low-income homeowners show 2021 values above 2007 levels. For the middle-income group, 50 out of 51 (98%) have recovered. The average Great Recession loss was largest for the middle-income group (-6.80%), followed by the low-income group (-4.35%), while the high-income group, as per the analysis, saw a slight average increase (+0.23%). COVID-era gains were broadly similar across groups, ranging from 14.1% (high income) to 17.4% (low income).

Table 1. State-Level Recovery Statistics by Income Group

Income Group	N	% Recovered	Avg. Δ 07–09	Avg. Δ 19–21
Low (bottom 20%)	51	100%	−4.35%	+17.4%
Middle (20–80%)	51	98%	−6.80%	+16.7%
High (top 20%)	51	100%	+0.23%	+14.1%

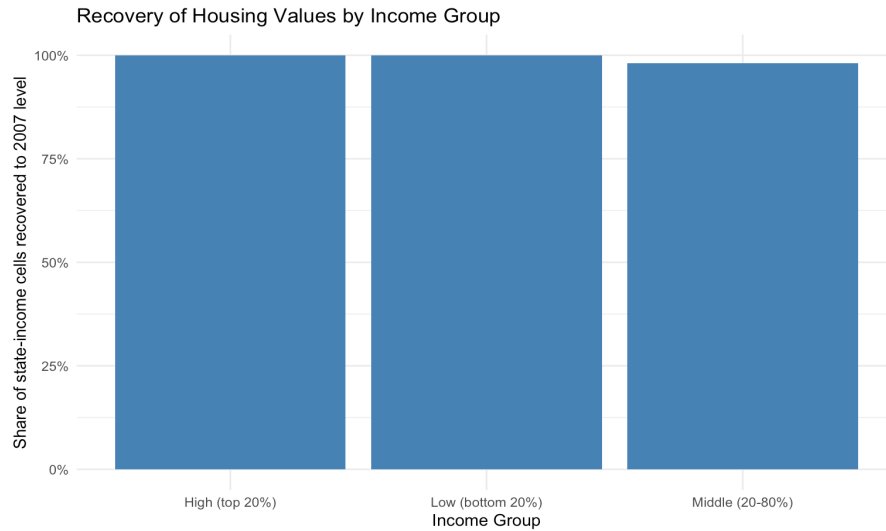


Figure 1. Share of state-income cells where 2021 housing values recovered to 2007 levels, by income group.

Figure 1 visualizes these recovery rates. The near-universal recovery in nominal terms is a positive finding but should be interpreted cautiously as it does not account for inflation, so the real (inflation-adjusted) recovery might be weaker and it reflects the average within each state, which may mask within-state heterogeneity.

5.2 Regression Results: Is There a Systematic Payback?

The more demanding test of the payback hypothesis is whether states that lost more during the Great Recession gained proportionally more during the COVID boom. Table 2 presents the linear regression results.

Table 2. Linear Regression Results: $\Delta Value(07-09)$ on $\Delta Value(19-21)$, by Income Group

Income Group	β	Std. Error	t-value	p-value	R ²
Low (bottom 20%)	-0.009	0.093	-0.100	0.921	0.000
Middle (20-80%)	-0.248	0.141	-1.760	0.085	0.059
High (top 20%)	-0.149	0.115	-1.294	0.202	0.033

No income group produces a statistically significant coefficient at conventional levels. The low-income group shows an essentially zero slope ($\beta = -0.009$, $p = 0.921$), indicating no relationship whatsoever between COVID-era gains and Great Recession losses. The middle-income group has the strongest result ($\beta = -0.248$, $p = 0.085$), which is marginally significant at the 10% level and explains about 6% of the variation. The high-income group falls in between ($\beta = -0.149$, $p = 0.202$). All R-squared values are very

low, indicating that COVID-era housing changes have minimal explanatory power for Great Recession losses at the state level.

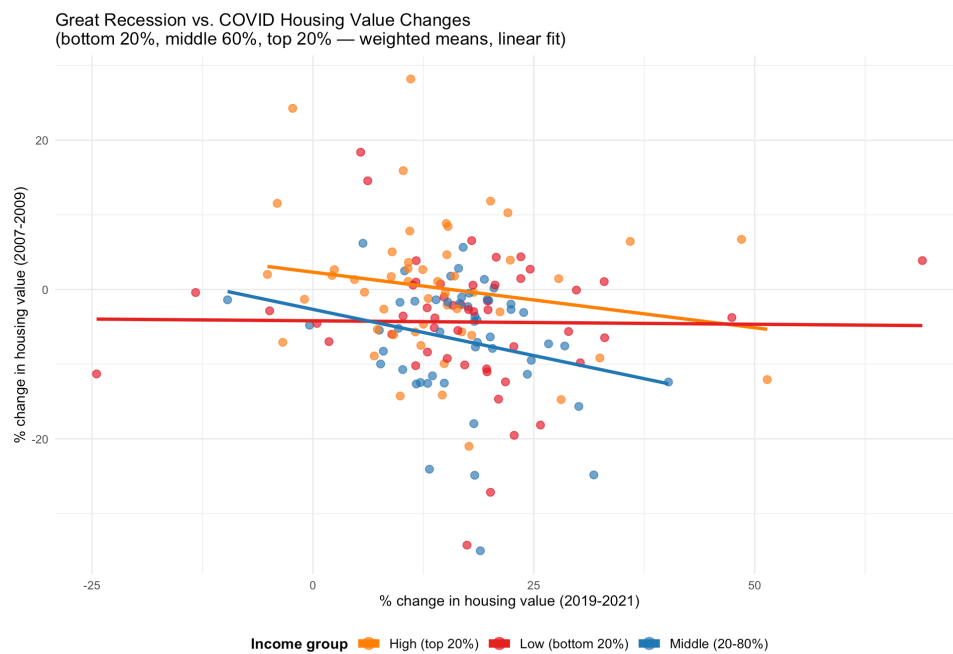


Figure 2. State-level scatter plot of Great Recession vs. COVID housing value changes, with linear regression lines by income group.

Figure 2 visualizes the state-level regressions. All three regression lines slope downward, which indicates payback. A downward slope means that states with more negative Great Recession changes tended to have slightly larger COVID gains. Especially for middle-income group, given the statistical significance at 10% level, there is an indication of housing value recovery.

5.3 PUMA-Level: Did Decimated Markets Recover?

Among the 1,076 PUMAs with complete data, 269 (25%) are classified as decimated, having experienced an average housing value decline of 16.9% between 2007 and 2009. The remaining 807 PUMAs (“other”) saw an average increase of 2.7% over the same period. During the COVID boom, both groups experienced comparable average gains; 16.2% for decimated PUMAs and 15.5% for others.

The key question is whether decimated PUMAs recovered to their pre-crisis levels. The answer is largely yes, but with a notable gap: 78.1% of decimated PUMAs had 2021 values at or above their 2007 levels, compared with 89.8% of non-decimated PUMAs. This 12-percentage-point gap indicates that while a substantial majority of even the hardest-hit markets nominally recovered, roughly one in five did not and these persistently depressed markets represent a meaningful share of local housing markets.

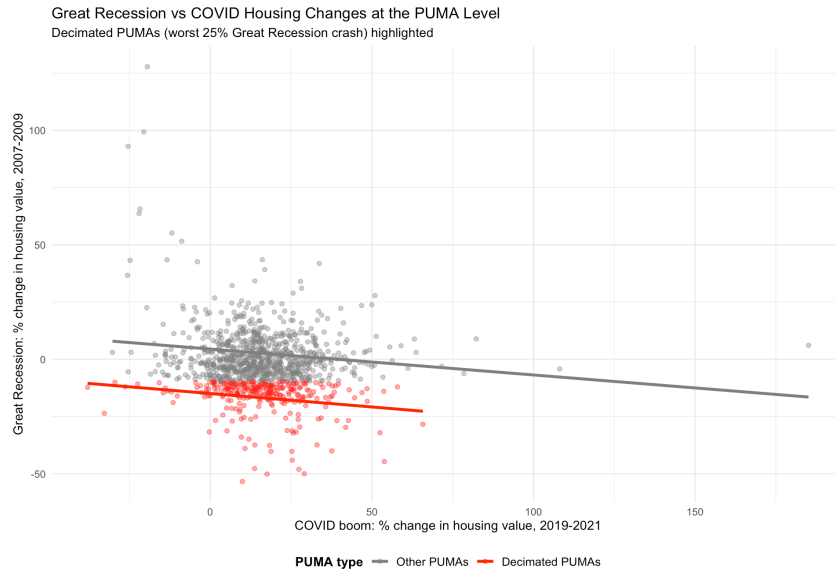


Figure 3. Scatter plot of Great Recession (2007–2009) vs. COVID boom (2019–2021) housing value changes at the PUMA level. Decimated PUMAs (worst 25% Great Recession crash) are shown in red.

Figure 3 shows the scatter plot of the two percentage changes at the PUMA level. Decimated PUMAs (red) are concentrated in the lower portion of the y-axis, reflecting their large Great Recession losses. The regression lines for both groups slope slightly downward, indicating a weak negative relationship: PUMAs with larger COVID gains did not systematically correspond to those with worse prior losses. The downward slope for decimated PUMAs is slightly steeper, but neither relationship is strong.

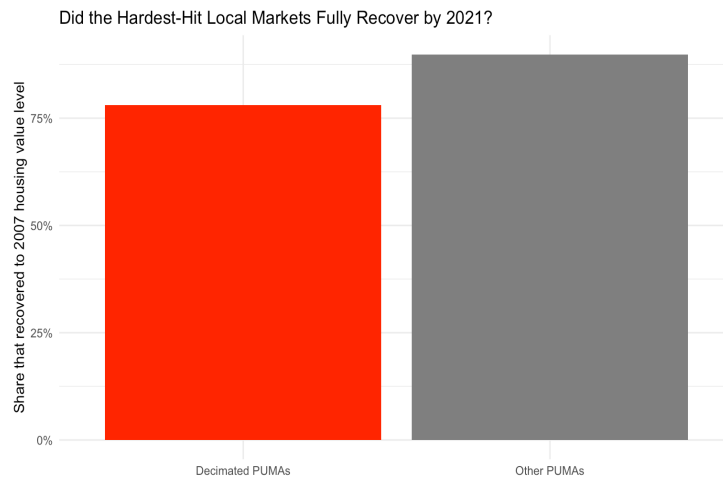


Figure 4. Share of PUMAs whose 2021 housing values recovered to or above 2007 levels, by decimation status.

Figure 4 presents the recovery rates directly. The gap between the two bars illustrates that the hardest-hit local markets were approximately 12 percentage points less likely to have fully recovered, even after the substantial COVID-era price increases. This finding suggests that the COVID boom was broadly beneficial but did not fully erase the geographic imprint of the Great Recession.

6. Discussion

The results tell a nuanced story about housing wealth recovery. On one hand, the nominal recovery is striking: across all income groups, every state-income cell for low- and high-income homeowners and 50 of 51 for middle-income homeowners show 2021 housing values above 2007 levels, and even among the most decimated local markets, more than three-quarters have nominally recovered. The COVID housing boom was broad-based and powerful enough to push most areas past their pre-crisis peaks in nominal terms.

On the other hand, the cross-sectional analysis reveals that this recovery was only weakly systematic. States that suffered larger Great Recession losses did not receive proportionally larger COVID-era gains, and the estimated “payback” slopes are small in magnitude, have very low explanatory power and are at best marginally significant. For low-income homeowners, the regression coefficients are statistically indistinguishable from zero; even in the best case, for the middle-income group, the negative slope is only significant at the 10% level and explains about 6% of the variation.

These findings are consistent with the broader themes of Professors Mian and Sufi’s work. The housing wealth destruction during the Great Recession was concentrated in specific geographic areas driven by local factors such as lending practices, speculation, and subprime exposure. The COVID boom, by contrast, was driven by a different set of national and local factors; which are interest rates, remote work, and supply constraints; that did not systematically overlap with the geography of the earlier bust. The result is that the two shocks are largely uncorrelated in their cross-sectional distribution.

Professor Campbell’s insight about portfolio composition provides a further lens. Because housing is the primary asset for middle and lower-income households, these groups were disproportionately exposed to the 2008 housing losses. The fact that the middle-income group shows the largest average decline (−6.80%) and the most suggestive regression slope suggests that this group was indeed most affected by housing cycles in both directions. Yet the absence of a strong, precisely estimated positive payback implies that sequential housing shocks do not clearly offset each other.

Zooming in to the PUMA level, the decimated markets experienced large Great Recession losses (−16.9% on average) and substantial COVID-era gains that were similar in size to those in other areas. A clear majority of these decimated PUMAs recovered to or above their 2007 levels by 2021, but their recovery rate (78.1%) lagged that of non-decimated PUMAs (89.8%) by roughly 12 percentage points. This gap underscores that even after an extraordinary boom, the hardest-hit local markets remain less likely to have fully recovered, and the geographic scars of the Great Recession have not been fully erased.

Several caveats are important. Firstly, all values are nominal; adjusting for inflation between 2007 and 2021 would reduce the apparent recovery. Secondly, the state-level analysis aggregates substantial within-state

heterogeneity. Thirdly, the state-level regressions are based on 51 observations per income group, which constrains statistical power and helps explain the low R-squared values. Lastly, using self-reported housing values from the ACS introduces measurement error, though this is a standard data source for housing research.

7. Conclusion

The results support a two-part answer to whether the COVID housing boom restored Great Recession housing losses. In nominal levels, the answer is largely yes: nearly all state income groups and a substantial majority of even the most severely hit local markets had 2021 housing values at or above their 2007 peaks. In terms of a systematic cross-sectional relationship, however, the evidence is weak. Areas and income groups that lost more in the Great Recession did not consistently experience larger COVID era gains, and the estimated payback slopes are small and explain little of the variation.

The policy implication is consistent with the themes of Professors Mian, Sufi and Campbell, which shows that asset price booms are an unreliable mechanism for repairing the distributional damage of asset price busts. If policymakers wish to address persistent wealth inequality arising from housing crises, they cannot assume that subsequent market recoveries will naturally undo past losses. Targeted interventions such as mortgage debt relief, homeownership assistance for lower and middle-income households, or policies that promote better portfolio diversification are needed to prevent the wealth effects of housing cycles from compounding across shocks and over time.

References

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